

Magnitude and Phase Aware Spectral Convolutional Neural Networks: Explainable Frequency-Domain Classification

Naga Kushal Vankadara¹[0009-0005-3799-5798], Gokaramaiah
Thota²[0009-0002-5048-8705], Nagaraju K¹[0000-0002-2358-2563], and Sathya
Babu Korra¹[0000-0002-5963-5735]

¹ Indian Institute of Information Technology Design and Manufacturing Kurnool¹
Jagannathagattu Hill, Kurnool-518008, Andhra Pradesh, India
122cs0015@iiitk.ac.in, knagaraju@iiitk.ac.in, ksb@iiitk.ac.in
² SRM University AP²
Neerukonda, Mangalagiri Mandal, Guntur District, Andhra Pradesh-522240, India
gokaramaiah.t@srmap.edu.in

Abstract. Convolutional neural networks (CNNs) have demonstrated strong performance in visual recognition tasks. However, conventional CNN architectures primarily operate on spatial-domain inputs, which may overlook important discriminative patterns and consequently limit model performance. In medical imaging, CNN models can also process frequency-domain representations obtained through the fast Fourier transform (FFT), which capture informative high-frequency features that are often less apparent in the spatial domain. Despite this advantage, interpretability in the frequency domain remains challenging because identifying the regions of attention in the original image space is non-trivial. In medical imaging applications, where transparent and trustworthy decision support is critical, this lack of explainability restricts clinical adoption. To address this challenge, this paper presents a spectral learning framework with explainability for interpreting frequency-domain CNN predictions. The input red, green, and blue (RGB) image channels are transformed using FFT to construct a 9-channel representation consisting of magnitude, cosine phase, and sine phase components. Although phase information provides localization-related cues, it is highly sensitive to perturbations and may introduce noise that degrades classification performance. To mitigate this issue, first-order gradients of the activation maps are fused with the activation maps before global average pooling. This integration stabilizes learning and enhances the recovery of discriminative spectral signatures. The proposed methodology is evaluated on both medical and non-medical image datasets using multiple CNN architectures to validate its robustness. Qualitative and quantitative results demonstrate the effectiveness of the proposed approach. Furthermore, the framework provides explainability for frequency-domain CNN models, making it particularly suitable for trustworthy medical imaging applications.

Keywords: Convolutional Neural Networks (CNNs) · Frequency-domain learning · Fast Fourier Transform (FFT) · Explainability · Class Activation Mapping (CAM) · Spectral Feature Representation · Spatial-Frequency Localisation

1 Introduction

Convolutional neural networks (CNNs) have significantly improved image classification performance across various tasks [2]. Models like ResNet-50 typically operate on spatial-domain inputs, analyzing raw pixel intensities [30]. However, spatial representations may miss important discriminative details [25]. In contrast, frequency-domain representations using the fast Fourier transform (FFT) capture high-frequency structural and textural patterns that are often overlooked by standard CNNs [4]. This is especially important in medical imaging, where subtle structural and texture variations are crucial for diagnosis.

An illustration of spatial and frequency-domain representations is shown in Fig. 1. Frequency-domain learning improves medical image analysis by enhancing feature representation and preserving important diagnostic details. Nahid et al. combined DFT and DCT features with residual CNNs (CNN-DF/CNN-DC) for breast cancer classification [13]. Ee Fey Goh et al. proposed FDCNN using RFFT-based operations to detect diabetic retinopathy efficiently while preserving fine details [4]. Kai Xu et al. used DCT representations with dynamic channel selection, highlighting CNN bias toward low-frequency components [30]. Totterstrom showed CNNs perform worse on pure Fourier spectra than RGB images [25], while Tang et al. introduced tKFC-Net, integrating FFT into U-Net for better segmentation and boundary detection [23]. Despite capturing high-frequency structural and textural patterns, frequency-domain methods lack interpretability, limiting understanding of model decisions.

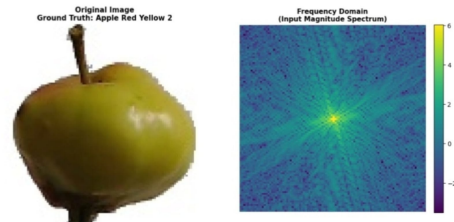


Fig. 1: Illustration of spatial and frequency domain representations of an image. The left panel shows the spatial domain image, while the right panel shows its corresponding frequency-domain representation obtained by using FFT.

Despite the representational benefits of frequency-domain learning, interpretability poses a non-trivial challenge [25] [30]. In the spectral domain, it is

not straightforward to identify which regions of the original image the model is attending to, since frequency components do not have a direct spatial correspondence. This limitation is especially consequential in medical imaging applications, where transparent and trustworthy decision support is not merely desirable but essential [7] [24] [15]. To address this, the framework incorporates a frequency-aware class activation mapping procedure. Salient spectral contributions are first identified in the frequency domain and subsequently mapped back to the spatial domain, producing class-discriminative attribution maps that localize diagnostically relevant regions with reduced visual artefacts [20].

Taken together, this paper presents a spectral learning framework that unifies frequency-domain inference, gradient-stabilised feature fusion, and spatially interpretable explanations. The resulting system not only improves classification performance over standard spatial baselines but also produces more reliable and interpretable outputs, making it well-suited for deployment in safety-critical domains such as medical image analysis.

2 Related Work

The spatial domain refers to the conventional representation of images where pixel values are arranged in their original two-dimensional layout, directly corresponding to physical locations—this is the standard input format for most CNN architectures. However, spatial-domain approaches have notable disadvantages: discriminative high-frequency patterns (like subtle edges or fine textures) are often less apparent in this representation, limiting both classification accuracy and the model’s ability to capture nuanced features critical for tasks like medical imaging.

The frequency domain represents images in terms of magnitude and phase components, exposing high-frequency patterns that remain hidden in spatial representations. The frequency domain offers advantages by revealing discriminative spectral signatures that enhance classification performance, while phase components inherently encode spatial localisation cues.

Many researchers have explored frequency-domain learning for medical imaging. Nahid et al. combined DFT and DCT features with residual CNNs and proposed CNN-DF/CNN-DC to study learning from spectral inputs [13]. Ee Fey Goh et al. developed FDCNN with RFFT-based layers and channel-independent convolution (CIC) to reduce complexity while preserving fine details [4]. Kai Xu et al. used DCT representations with dynamic channel selection, revealing CNN bias toward low-frequency components [30]. Totterstrom showed that CNNs perform worse on pure Fourier inputs than RGB images [25]. Tang et al. proposed tKFC-Net, integrating FFT into U-Net with twin-kernel Fourier convolution (t-KFC) to capture semantic context and boundary details [23].

Explainable artificial intelligence (XAI) focuses on understanding how complex models make decisions and predictions [7]. Common XAI techniques include LIME and SHAP. LIME is a post-hoc method that explains individual predictions by approximating a black-box model with a simple interpretable model

(e.g., linear) around a specific instance [17]. For images, LIME segments the input into superpixels, creates perturbed samples by masking them, weights them based on similarity, and fits a linear model to estimate each superpixel’s contribution [7] [17].

SHAP is a model-agnostic explanation framework that assigns each feature an importance value for a prediction using Shapley values from game theory, showing its contribution to the model output [8]. For images, it estimates the importance of pixels or pixel groups. SHAP satisfies three properties: local accuracy, missingness, and consistency. It computes feature attributions using methods like Kernel SHAP and Deep SHAP [7] [8].

Although LIME and SHAP are effective, they have key limitations for image explainability. LIME lacks precise spatial localisation, as superpixel masks vary and do not capture exact boundaries [7]. It is also computationally expensive, taking around 10 minutes per image for deep models like Inception [17]. SHAP is computationally intractable for high-dimensional images like $224 \times 224 \times 3$, requiring evaluation of 2^M subsets (NP-hard with $O(2^M + M^3)$ complexity) [8]. Additionally, SHAP assumes feature independence, leading to unrealistic pixel combinations and less reliable attributions [7] [8].

To overcome the limitations of LIME and SHAP, CAM (Class Activation Map) provides a more efficient XAI approach for image explainability. It offers precise spatial localisation by using convolutional feature maps to highlight important regions and fine details [20]. It also reduces computational cost by requiring only a single forward and partial backward pass, avoiding heavy perturbation sampling [20]. Additionally, it is broadly applicable without needing architectural modifications [1].

Even though CAM is effective compared to LIME and SHAP, there are a few disadvantages of CAM that make it difficult to operate in the frequency domain, such as loss of high-frequency information, noisy and unstable explanations, inability to provide frequency-specific explanations, and architectural constraints and computational overhead.

Although frequency-domain methods in deep learning have improved performance and efficiency, they mainly focus on classification, segmentation, and acceleration, without addressing explainability. Despite their advantages over spatial-domain approaches, explainability in the frequency domain remains unexplored. This paper introduces frequency-domain explainability using CAM, enabling visualisation of important frequencies and mapping them back to the spatial domain to identify the pixel regions influencing the model’s decision.

3 Methodology

The proposed framework transforms spatial-domain images into the frequency domain using the FFT, which decomposes each input into magnitude and phase components. The phase encodes critical spatial localisation information but is inherently sensitive to noise and discontinuities. To address this, each RGB channel is represented using log-magnitude along with sine and cosine transforma-

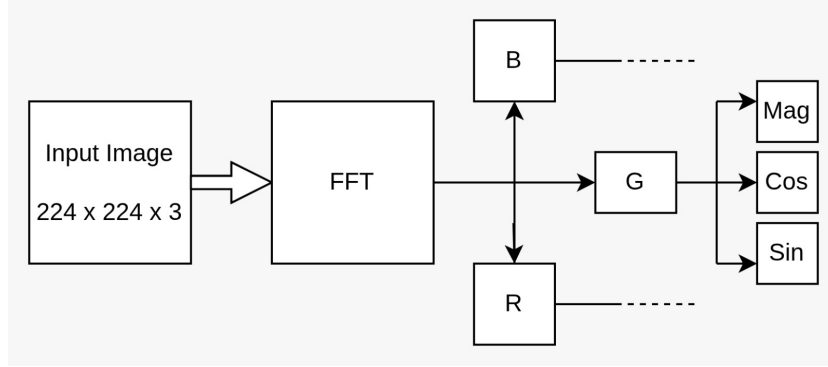


Fig. 2: Each RGB channel is decomposed in the frequency domain into magnitude, cosine-phase, and sine-phase components (illustrated for the G channel), forming a nine-channel spectral representation that serves as input to the CNN for improved feature robustness and localisation.

tions of the phase, ensuring a stable and continuous encoding. This results in a nine-channel representation (3 channels $\times$ magnitude, $\cos \phi$, $\sin \phi$), effectively preserving both spectral characteristics and structural details for improved feature learning. The overall proposed system diagram is shown in Figure 2.

3.1 Spatial-to-frequency domain conversion

An RGB image, which has three colour channels, is transformed into the frequency domain. This is done by applying the 2D discrete Fourier transform separately to each colour channel, resulting in a complex spectrum. The spectrum is then adjusted so that the lower frequency parts are in the middle using a method called 'fftshift'. Once this is done, the spectrum is split into two parts: magnitude and phase. The magnitude, $M_c(u, v)$, shows how much of each frequency is present, which relates to texture and edges in the image. The phase, $\Phi_c(u, v)$, keeps the spatial arrangement of the image intact.

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad (1)$$

where u and v represent the spatial frequency coordinates [19]. The resulting output $F(u, v)$ is complex-valued, consisting of a real part (Re) and an imaginary part (Im). [2] [19]

$$M_c(u, v) = |F(k, l)| = \sqrt{Re(F(k, l))^2 + Im(F(k, l))^2} \quad (2)$$

Here, (k, l) are the frequency positions. [25]

$$\Phi_c(u, v) = \arg(F(k, l)) = \arctan \left(\frac{Im(F(k, l))}{Re(F(k, l))} \right) \quad (3)$$

To deal with the wide range of values in the magnitude spectrum, a logarithmic compression method is used along with some numerical adjustments. This is done using:

$$L_c(u, v) = \ln(\max(M_c(u, v), \epsilon) + \epsilon) \quad (4)$$

This helps make the data more consistent for learning. Since phase values repeat and can have sudden changes, they are converted using sine and cosine functions:

$$\begin{aligned} P_c^{\cos}(u, v) &= \cos(\Phi_c(u, v)), \\ P_c^{\sin}(u, v) &= \sin(\Phi_c(u, v)) \end{aligned} \quad (5)$$

This creates a smooth and controlled representation. The processed log-magnitude and phase data from three different channels are then combined into a single nine-channel frequency tensor

$$\mathbf{X}_{\text{freq}} \in \mathbb{R}^{9 \times H \times W}$$

. This format keeps both the energy distribution and the spatial details, making it suitable for further processing with convolutional methods.

3.2 CNN architecture

The architecture used is a modified version of a CNN model that works with 9-channel frequency-domain inputs. The original CNN model is designed for three-channel RGB images, so the first convolutional layer is changed such that it takes 9 channels and produces l channels, with half the height and width of the input. It uses a 7x7 kernel, a stride of 2, and padding of 3.

$$\text{Conv1} : \mathbb{R}^{9 \times H \times W} \rightarrow \mathbb{R}^{l \times \frac{H}{2} \times \frac{W}{2}}$$

To use the pre-trained weights from ImageNet, a transfer-and-scale method is applied. The first three channels of the new layer are set to the pre-trained weights, while the remaining six channels are initialised using a scaled version of the Kaiming He initialisation. This setup keeps the useful spatial features from the pre-trained model while allowing the model to learn frequency-domain information gradually.

$$W_{\text{conv1}}[:, 0:3, :, :] = W_{\text{pretrained}}, \quad (6)$$

$$W_{\text{conv1}}[:, 3:9, :, :] = 0.1 \cdot W_{\text{Kaiming}} \quad (7)$$

After the modified input layer, features are taken out using the standard CNN residual blocks (layers 1 to 4). Each block does the following:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, W_i) + \mathbf{x} \quad (8)$$

which $\mathcal{F}$ includes steps like convolution, batch normalisation, and applying the ReLU function. The last layer (layer 4) gives out deep feature maps:

$$\mathbf{A} \in \mathbb{R}^{B \times C_{\text{backbone}} \times h \times w}$$

where B is the batch size, C_{backbone} is the number of feature channels produced from the last convolutional layer, and $h \times w$ denotes the spatial resolution of the feature maps. These feature maps are used in the next steps for combining information from two different domains and for making the model's decisions easier to understand.

3.3 Dual-frequency feature fusion

The dual-frequency feature fusion module works with activation maps $\mathbf{A} \in \mathbb{R}^{B \times C \times h \times w}$ that are taken from the last residual block. To get more information about the spectral details in these spatial features, a second-level 2D FFT is done separately for each channel.

$$G_k(u, v) = \text{FFT2D}(A_k(x, y)), \quad M_k^{(2)} = |G_k|. \quad (9)$$

This gives magnitude spectra, which are then log-scaled and normalised for each channel to create frequency-domain feature maps $\mathbf{A}_{\text{freq}}$. This process helps reveal patterns like repeating structures, main spatial scales, and periodic elements that aren't easily seen in the original spatial data, making the representation richer with more complex structural details.

Because spatial and frequency features exist in different numerical ranges, separate batch normalisation layers are used to match their distributions

$$\tilde{\mathbf{A}} = \text{BN}_{\text{spatial}}(\mathbf{A}), \quad \tilde{\mathbf{A}}_{\text{freq}} = \text{BN}_{\text{freq}}(\mathbf{A}_{\text{freq}}) \quad (10)$$

$$\mathbf{C} = [\tilde{\mathbf{A}}; \tilde{\mathbf{A}}_{\text{freq}}] \in \mathbb{R}^{B \times 2C \times h \times w} \quad (11)$$

The normalised features are combined along the channel dimension and then go through a channel attention mechanism that uses global average pooling.

$$\mathbf{z} = \text{GAP}([\tilde{\mathbf{A}}; \tilde{\mathbf{A}}_{\text{freq}}]), \quad \mathbf{s} = \sigma(W_2, \delta(W_1 \mathbf{z})). \quad (12)$$

where GAP denotes global average pooling, δ is the ReLU activation function, σ is the sigmoid activation function, $W_1 \in \mathbb{R}^{C \times 2C}$ is the dimensionality reduction matrix, and $W_2 \in \mathbb{R}^{2C \times C}$ is the expansion matrix.

$$\hat{\mathbf{C}} = \mathbf{s} \odot \mathbf{C} \quad (13)$$

This attention process helps the model focus on important spatial and frequency channels while reducing the impact of less useful ones. It learns how different channels relate to each other across both domains.

Finally, the weighted features are split and merged using a learnable fusion method:

$$\begin{aligned} \hat{\mathbf{A}}_s &= \hat{\mathbf{C}}[:, :, C, :, :], \quad \hat{\mathbf{A}}_f = \hat{\mathbf{C}}[:, C :, :, :] \\ \mathbf{A}_{\text{fused}} &= \alpha \cdot \hat{\mathbf{A}}_s + \beta \cdot \hat{\mathbf{A}}_f \end{aligned} \quad (14)$$

where $\alpha = \sigma(\alpha \cdot \text{raw})$ and $\beta = \sigma(\beta \cdot \text{raw})$ are trainable values between 0 and 1. Initially, more weight is given to spatial features due to their better early performance. As training progresses, the model adjusts the importance of frequency information. The final feature combines local spatial details and global spectral patterns before average pooling and classification.

The combined feature maps $\mathbf{A}_{\text{fused}} \in \mathbb{R}^{B \times C_{\text{backbone}} \times h \times w}$ are first turned into a fixed-length form using global average pooling. This process creates two vectors:

$$\mathbf{v} = \frac{1}{h \cdot w} \sum_{i=1}^h \sum_{j=1}^w, \quad \mathbf{A}_{\text{fused}}(:, :, i, j) \in \mathbb{R}^{B \times C_{\text{backbone}}} \quad (15)$$

which gather spatial information while making the model less sensitive to changes in space.

This pooled vector v is then sent through a three-layer fully connected classification part as shown.

$$\begin{aligned} \mathbf{v} &\xrightarrow{\text{Dropout}(p_1)} \xrightarrow{W_1 \in \mathbb{R}^{C_{\text{backbone}} \times C_1}} \xrightarrow{\text{ReLU}} \xrightarrow{\text{BN}} \\ &\xrightarrow{\text{Dropout}(p_2)} \xrightarrow{W_2 \in \mathbb{R}^{C_1 \times C_2}} \xrightarrow{\text{ReLU}} \xrightarrow{\text{BN}} \\ &\xrightarrow{\text{Dropout}(p_3)} \xrightarrow{W_3 \in \mathbb{R}^{C_2 \times K}} \mathbf{o} \end{aligned}$$

The output is a vector $\mathbf{o} \in \mathbb{R}^{B \times K}$ that gives the predicted scores for K classes. Using ReLU activation along with batch normalisation helps the model learn better and train faster. The dropout rates start at p_1 , then go down to p_2 , and finally to p_3 . This setup applies more regularisation in the early layers and less in the later ones, helping to prevent overfitting while still allowing the model to capture enough details for accurate predictions.

3.4 Explainability

Class activation mapping (CAM) variants have Grad-CAM [20], XGrad-CAM [3], Smooth Grad-CAM [14], Score-CAM [29], Ablation-CAM [16], Eigen-CAM [12] and Grad-CAM++ [1]. In this sub-section, Score-CAM is implemented in the frequency domain; the same procedure can be applied to other CAM-based methods. Score-CAM works on the combined activation maps $\mathbf{A}_{\text{fused}} \in \mathbb{R}^{1 \times C \times h \times w}$, which are created before the global pooling step. For each input, each channel is scaled up to the original image size using a bilinear method. Then, these channels are adjusted to fit a range from 0 to 1. This makes them suitable as soft importance maps that show where the model focuses its attention in space.

$$\bar{A}_k = \frac{A_k^\uparrow - \min(A_k^\uparrow)}{\max(A_k^\uparrow) - \min(A_k^\uparrow)} \quad (16)$$

where it $A_k^\uparrow$ is called an unsampled activation map. The normalised map, $\bar{A}_k$, acts as a soft spatial mask. This mask is applied to the frequency-domain input,

resulting in $\mathbf{X} * k = \mathbf{X} * \text{freq} \odot \bar{A}_k$. This process helps keep the areas that are most important according to the (k)-th fused feature.

Each of these masked inputs goes through the whole network to produce a score that shows how confident the model is about each class.

$$s_k = \text{softmax}((f(\mathbf{X}_k)) * t) \quad (17)$$

where $f(\cdot)$ denotes the model prediction function and t is the target class index.

Finally, the class activation map is created by combining the original fused feature maps, using the confidence scores as weights, and then applying ReLU activation to keep only the areas that contribute positively.

$$\text{CAM}(u, v) = \text{ReLU}\left(\sum_{k=1}^C s_k \cdot A_k(u, v)\right) \quad (18)$$

This keeps the original Score-CAM idea and makes sure the explanations are clearly connected to the spatial and frequency patterns that the DFFF learns together.

3.5 Frequency to spatial domain mapping

The CAM created in the frequency domain is converted back to the spatial domain so people can easily understand it. The CAM acts as a mask that multiplies the frequency magnitude. When combined with the original phase, the inverse FFT creates a saliency signal in the spatial domain.

$$I_{\text{sal}} = \text{IFFT2D}(\text{CAM} \cdot M \cdot e^{j\Phi}) \quad (19)$$

The final saliency map is made by averaging across all channels. It is then flipped because areas with high energy in the frequency domain are related to edges and textures, not the centres of objects.

Now, a Gaussian filter with a standard deviation of 2.5 is used to smooth the map with a threshold based on the 40th percentile. To make it cleaner, it goes through two steps: first, a closing operation using a 5x5 structuring element, and then an opening operation using a 3x3 structuring element. Finally, the contrast is increased using a power-law transformation with an exponent of 0.7.

The process to bring back spatial images from the frequency domain undoes the steps that were taken. With the log-magnitude representation L and a phase Φ , the complex spectrum can be reconstructed as

$$\hat{F}(u, v) = e^{\text{clamp}(L, -10, 10)} \cdot e^{j\Phi(u, v)} \quad (20)$$

The spectrum is then shifted back using inverse fftshift, and the spatial image is created by applying the inverse FFT.

$$I_{\text{reconstructed}}(x, y) = \text{Re}\left(\text{IFFT2D}\left(\text{ifftshift}\left(\hat{F}(u, v)\right)\right)\right) \quad (21)$$

4 Experimentation and Results

4.1 Datasets used

This paper uses two publicly available datasets: Fruits-360 dataset and the Brain Tumor MRI dataset. The Fruits-360 dataset is a large-scale image classification dataset available on Kaggle. It consists of RGB images of fruits and vegetables, each of size 100×100 pixels. The dataset contains a total of 130,344 images, divided into 110,792 training samples, 19,552 validation samples, and a separate test set of 43442 images. The Brain Tumor MRI dataset consists of labelled MRI scans used for multi-class brain tumor classification, including categories such as glioma, meningioma, and pituitary tumors. It contains a total of 5,600 images in the training set, which is further divided into 4,760 training samples and 840 validation samples, along with a separate test set of 1,600 images.

4.2 Pre-processing steps

Training set During training, input images go through a stochastic augmentation pipeline which includes random horizontal flipping with probability 0.5, random vertical flipping with probability 0.3, random rotation within ± 20 degrees, colour jittering applied to brightness (± 0.2), contrast (± 0.2), saturation (± 0.2), and hue (± 0.1), and random affine transformations with up to 10% translation and scaling between 0.9 and 1.1. The pixel values are turned into tensors that range from 0 to 1, and then they are adjusted using specific average and standard deviation values from the ImageNet dataset: mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

Testing set The test set undergoes only the deterministic steps: no data augmentation is applied, but it undergoes tensor conversion, ImageNet normalisation (the same as the training set) and FFT-based frequency domain conversion.

4.3 Training strategy

Loss function The model is trained using cross-entropy loss with label smoothing. The loss function is:

$$\mathcal{L} = - \sum_{k=1}^K \tilde{y}_k \log(\hat{y}_k) \quad (22)$$

where the smoothed labels are defined as:

$$\tilde{y}_k = (1 - \varepsilon) y_k + \frac{\varepsilon}{K}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{o})$$

where $\mathbf{y}$ one-hot encoded ground truth, $\hat{\mathbf{y}}$ is predicted probability. ε is a smoothing parameter with a value of 0.1. This prevents the model from becoming overconfident and improves generalisation.

Optimiser and learning rate The AdamW optimiser is used with decoupled weight decay $\lambda = 5 \times 10^{-4}$. The learning rate approach used is the pre-trained CNN model, which is adjusted with a small learning rate $\eta_{\text{pre}} = 10^{-5}$, while the new parts added to the model, like the modified input convolution, the DFFF module, and the classification head, are trained with a larger learning rate $\eta_{\text{new}} = 5 \times 10^{-4}$. The learning rate follows a cosine annealing schedule with warm restarts:

$$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{T_{\text{cur}}}{T_i}\pi\right) \right) \quad (23)$$

where $T_0 = 10$ is the initial restart period, $T_{\text{mult}} = 2$ the period multiplier, and $\eta_{\min} = 10^{-7}$. This schedule lets the optimiser break free from local minima by increasing the learning rate at regular intervals.

Mixed Precision Training Training uses automatic mixed precision (AMP), where the main calculations happen with float16, but the updates to the model’s settings are done with float32. The system automatically adjusts the scaling of gradients to make sure they stay accurate when using half-precision numbers.

Gradient clipping To stop the gradients from getting too large, which can be a problem because the network uses FFT operations, we use a technique called global gradient norm clipping. This sets a maximum limit on how big the gradients can be, which is 0.5:

$$\mathbf{g} \leftarrow \frac{0.5}{\|\mathbf{g}\|_2} \cdot \mathbf{g} \quad \text{if } \|\mathbf{g}\|_2 > 0.5 \quad (24)$$

4.4 Setup

The code is designed for a GPU (preferably 16 GB) with *CUDA*, using *PyTorch* and *torchvision*. *SciPy* handles saliency post-processing, and *Matplotlib* is used for visualisation. CuDNN benchmark mode is enabled, and deterministic mode is disabled for faster training. With CUDA, mixed-precision training using AMP and *GradScaler* improves speed and reduces memory usage. The data are split into 85% training and 15% validation (seed = 42), and dataloaders use a batch size of 128, pin memory, and a prefetch factor of 2.

4.5 Hyperparameters

The model was trained for 50 epochs with a batch size of 128 and an initial learning rate of 0.001. Different learning rates were used for pre-trained layers ($1e^{-5}$) and newly added layers ($5e^{-4}$). Dropout was applied in the classifier to prevent overfitting. Early stopping was used to halt training if validation performance did not improve for 15 epochs.

Table 1: Comparative performance of CNN backbones with the proposed dual-frequency feature fusion framework on the Fruits-360 test set.

Metric	ResNet-50	MobileNetV2	DenseNet-121	EfficientNet-B0	VGG-19
Overall Accuracy (%)	90.39	89.94	89.65	87.78	86.51
Cohen’s Kappa Score (%)	90.34	89.88	89.60	87.71	86.45
Overall Precision (%)	92.25	90.75	91.54	91.35	89.27
Overall Recall (%)	90.48	89.89	89.90	87.82	86.71
Overall F1 Score (%)	90.30	89.73	89.79	87.72	86.40
Overall Specificity (%)	99.96	99.94	99.95	99.93	99.94
Overall Error Rate (%)	9.52	0.10	0.08	16.18	14.48

4.6 Quantitative results analysis

Table 1 and Table 2 presents the quantitative performance of the proposed dual-frequency feature fusion framework evaluated across five CNN backbone architectures: ResNet-50 [5], MobileNetV2 [18], DenseNet-121 [6], EfficientNet-B0 [22], and VGG-19 [21], on the Fruits-360 and Brain Tumor MRI datasets, respectively. The selected architectures cover a wide range of designs from deep and dense networks to lightweight and scaled models, ensuring robust validation of the framework across different model types and capacities. These results show

Table 2: Comparative performance of CNN backbones with the proposed dual-frequency feature fusion framework on the Brain Tumor MRI test set.

Metric	ResNet-50	MobileNetV2	DenseNet-121	EfficientNet-B0	VGG-19
Overall Accuracy (%)	85.44	86.78	87.62	83.51	86.09
Cohen’s Kappa Score (%)	82.65	85.10	85.93	81.74	83.91
Overall Precision (%)	87.32	88.75	89.25	86.35	87.89
Overall Recall (%)	85.57	86.46	87.81	85.43	86.28
Overall F1 Score (%)	85.50	86.37	87.73	85.26	86.14
Overall Specificity (%)	95.86	96.04	97.05	94.59	95.97
Overall Error Rate (%)	14.27	12.26	11.08	16.38	13.05

the benefit of this approach compared to pure spatial domain CNNs that only use pixel values and have trouble recognising repeating patterns, textures, and edges. The new method solves this by using a 9-channel frequency format based on FFT, which includes both the strength and the timing of the signals, giving extra information that goes beyond standard RGB inputs. The consistently high specificity and precision across both datasets and all five architectures indicate that the dual-frequency input reduces model confusion in regions where spatial features are ambiguous, a common limitation of spatial-domain CNNs in fine-grained and medical image classification.

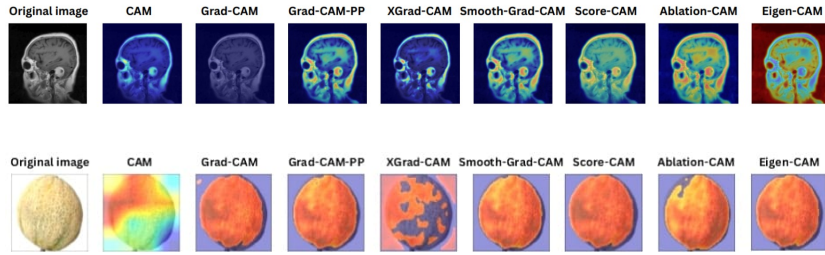


Fig. 3: Qualitative comparison of frequency-domain heatmaps with class activation mapping methods.

4.7 Qualitative results analysis

Figure 3 illustrates the activation maps generated by eight different Class Activation Mapping (CAM) variants applied to a cantaloupe sample from both datasets. Each variant offers distinct visualization characteristics that reveal how the dual-frequency feature fusion model interprets frequency-enriched features during classification. Spatial-domain CAM visualizations frequently produce diffuse, poorly localized heatmaps that spread activation across broad image regions rather than clinically meaningful structures, making them difficult to interpret and less useful for medical decision support.

The proposed spectral learning framework addresses this limitation by providing structured frequency-domain information, enabling activation maps to align with real anatomical and pathological features. Its precise localisation comes from using magnitude, cosine phase, and sine phase components, which capture high-frequency boundary and texture details often suppressed in spatial representations.

5 Conclusion and Future Work

This paper proposes a nine-channel spectral representation in the frequency domain. Each RGB channel is decomposed into one magnitude component and two phase components (sine and cosine). The sine-cosine formulation improves localisation robustness by preserving continuous phase information. However, explicit phase encoding may introduce noise that degrades performance. To mitigate this, second-order gradients from the final convolutional feature maps are fused with the features using an attention-based dual-frequency fusion module. Training stability is maintained through NaN/Inf safeguards, mixed-precision training, and gradient clipping. CAM methods evaluated on a fruit dataset highlight frequency-informed regions, showing that spectral cues complement spatial features with minimal computational overhead.

Future work will introduce a multi-resolution frequency pyramid (100×100 , 50×50 , 25×25) with lightweight CNN processing and attention-based fusion. A multi-scale CAM framework will provide frequency-aware interpretability.

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